Circuit Connections

Below is the **circuit connection table for the current Smart Pill Dispenser setup**, using the GPIO assignments present in the project code. The motor driver used in the code is **TB6612FNG**.

Raspberry Pi 5 → TB6612FNG Motor Driver

Raspberry Pi 5	BCM GPIO	TB6612FNG Pin	Function
GPIO 17	17	AIN1	Motor 1 direction
GPIO 27	27	AIN2	Motor 1 direction
GPIO 18	18	PWMA	Motor 1 speed/PWM
GPIO 22	22	BIN1	Motor 2 direction
GPIO 23	23	BIN2	Motor 2 direction
GPIO 13	13	PWMB	Motor 2 speed/PWM
GPIO 24	24	STBY	Driver enable/standby
GND	—	GND	Common ground

These GPIO assignments are directly defined in the application.

TB6612FNG → Motors

TB6612FNG Pin	Connected To	Purpose
A01 / AO1	Motor 1 terminal	Motor 1 output
A02 / AO2	Motor 1 terminal	Motor 1 output
B01 / BO1	Motor 2 terminal	Motor 2 output
B02 / BO2	Motor 2 terminal	Motor 2 output

Motor 1 = Pill 1 and **Motor 2 = Pill 2** in the software.

Raspberry Pi 5 → Buzzer

Raspberry Pi 5	BCM GPIO	Buzzer Connection	Function
GPIO 26	26	Buzzer + / signal	Buzzer control
GND	—	Buzzer −	Ground

The application defines `BUZZER_PIN = 26` and drives it HIGH/LOW for the alert pattern.

TB6612FNG Power Connections

TB6612FNG Pin	Connect To	Purpose
VM	3.7V Battery +ve	Motor power
VCC	Raspberry Pi 3.3V	Logic power
GND	3.7V Battery -ve & RPi GND	Common ground
STBY	Raspberry Pi GPIO 24	Driver enable

Cameras

Component	Connection	Purpose
Patient Camera	Raspberry Pi CSI camera connector	Patient face/hand/mouth monitoring
Pill Camera	Raspberry Pi CSI camera connector	Tray and pill detection

Important: The GPIO numbers above are **BCM GPIO numbers**, matching `GPIO.setmode(GPIO.BCM)` in the application.